

# Duolingo Vocabulary Retention: Optimising the Spaced-Repetition Review Window

A data-driven investigation into when Duolingo should re-show a learned word — balancing forgetting risk against over-practice — using 499,000+ practice sessions across 6 languages and 8 course pairings.

## PRIMARY KPI

Word Recall Rate

## METHOD

11 Hypotheses · Python (pandas, scipy, statsmodels)

## DATASETS

4 cleaned tables

**499,436**

Learning traces  
analysed

**75,294**

Unique learners

**6,211**

Unique words scored

**90.45%**

Overall word recall rate

**8**

Language course  
pairings

# Executive Summary

Duolingo operates one of the world's largest language-learning platforms. A core part of the product experience is spaced repetition — deciding when to re-show a previously learned word so it is neither forgotten nor over-practiced. This analysis of 499,436 practice traces, 6,211 scored words and 8 course pairings identifies exactly where recall breaks down, and what to change in the review schedule.

## 90.45%

**Overall Word Recall Rate**  
822,036 correct / 908,795 attempts

## 87.9%

**Recall at 3+ Months Lag**  
↓ 3.84pp from the <1 hour peak

## ≈10.1

**Optimal Exposure Turning Point**  
Model-adjusted for lag & language

## 84% vs 94%

**Hardest vs Easiest Word Quartile**  
 $r = 0.739$  difficulty–recall correlation

### Six Findings That Matter Most

- **Recall decays fastest in the first 6 hours** — 91.7% → 90.2%, a 1.49pp single-step drop, the steepest of the whole curve.
- **Exposure count stops helping around 10.1 repetitions** — raw recall peaks near 10–19 exposures then declines; the model-adjusted curve confirms a turning point at 10.1.
- **Word difficulty is the strongest single predictor of recall** ( $r = 0.739$ ,  $p < 0.0001$ ) — a 9.6pp gap separates the hardest and easiest quartiles.
- **French is a structural outlier** — lowest course recall (89.30%), longest average lag, and the fastest 90%-threshold breach of any language.
- **Function words underperform content words everywhere** — Auxiliary verbs recall at 80.9%, vs 94.9% for interjections and ~90.7% for adjectives/nouns.
- **Lag and exposure interact** — at 3+ months lag, 10–19 prior exposures retain 88.6% vs 86.9% at 1–2 exposures, a 1.7pp protective buffer.

### Business Problem

After a learner has learned a word, how long can Duolingo wait before reviewing it without the learner forgetting it? The goal is to identify where recall starts becoming weak, and which learner or word characteristics require earlier or more frequent review.

### Top 3 Recommendations

- **1.** Set language-specific initial review intervals (French shortest, Portuguese/Italian longest) — see Section 11.
- **2.** Give the hardest word quartile shorter intervals and a format-change trigger past 10.1 exposures.
- **3.** Replace raw exposure count with rolling recall accuracy as the mastery signal.

Business Understanding & Data Understanding

Business Context

Duolingo operates one of the world's largest language learning platforms, serving hundreds of millions of learners through short, gamified sessions. A core part of the product is spaced repetition — deciding when to re-show a learned word so it is neither forgotten nor over-practiced.

OBJECTIVE & KEY METRIC

Find where recall weakens and which characteristics need earlier review. Primary KPI — **Word Recall Rate = Correct Recalls ÷ Recall Attempts.**

Supporting KPIs & Business Dimensions

Supporting KPIs

- Avg Session Recall
- Item Recall Rate
- Recall by Lag / Exposure Bin
- Word Difficulty & Course Recall

Business Dimensions

- Time (lag)
- Exposure
- Word / Difficulty / POS
- Language / Course / Learner behaviour

Data Understanding — What One Row Represents, Per Dataset

| Dataset                  | Grain (one row ⇒)                                     | Rows    | Cols | Key columns                                          |
|--------------------------|-------------------------------------------------------|---------|------|------------------------------------------------------|
| traces_cleaned           | One learner × one word × one practice observation     | 499,436 | 15   | learning_language, lag_days, lag_bin, history_seen   |
| word_difficulty_cleaned  | One word × one learning language (aggregated)         | 6,211   | 12   | surface_form, pos, item_recall_rate, difficulty_rank |
| forgetting_curve_cleaned | One language × lag × exposure bucket (pre-aggregated) | 310     | 10   | lag_bin, practice_bin, avg_session_recall, n_traces  |
| courses_cleaned          | One UI-language → learning-language course pairing    | 8       | 11   | course, ui_language, item_recall_rate, avg_lag_days  |

Business Questions Driving the Analysis

Q1 — DECAY

Where is the critical recall drop-off as time since practice grows?

Q2 — PRACTICE HISTORY

Does more practice reliably help, or do returns diminish?

Q3 — WORD DIFFICULTY

How wide is the recall gap between hard and easy words?

Q4 — COURSE PERFORMANCE

Which language courses underperform on recall, and why?

Q5 — GRAMMAR CATEGORIES

Do parts of speech show systematically different recall?

Q6 — INTERACTION SIGNAL

Should scheduling combine lag and exposure signals?

# Data Quality & Cleaning

Every check below was run against the actual cleaned datasets. All four tables came back clean: zero duplicate rows, zero invalid numeric values, zero out-of-range probabilities. The only meaningful gap — grammar\_tags — was deliberately retained rather than dropped, since it is not required to calculate Word Recall Rate.

| Dataset                  | Rows    | Cols | Duplicates | Missing values               | Notes                                             |
|--------------------------|---------|------|------------|------------------------------|---------------------------------------------------|
| traces_cleaned           | 499,436 | 15   | 0          | grammar_tags: 50,956 (10.2%) | 0 negative / invalid values of any kind           |
| word_difficulty_cleaned  | 6,211   | 12   | 0          | grammar_tags: 754 (12.1%)    | n_traces range 100–123,099 (retained, not capped) |
| forgetting_curve_cleaned | 310     | 10   | 0          | None                         | 9 lag bins × 5 exposure bins                      |
| courses_cleaned          | 8       | 11   | 0          | None                         | Exactly 8 course pairings, all populated          |

Cleaning Checks & Decisions (traces\_cleaned)

| Check                                             | Result              | Decision                                                    |
|---------------------------------------------------|---------------------|-------------------------------------------------------------|
| Rows (traces_cleaned)                             | 499,436             | Keep                                                        |
| Duplicate rows (all 4 datasets)                   | 0                   | Nothing to remove                                           |
| Missing grammar_tags                              | 50,956 (10.2%)      | Investigate, don't drop — not required for Word Recall Rate |
| Negative lag_days / history_seen / session values | 0                   | Good — no correction needed                                 |
| history_correct > history_seen                    | 0                   | Good — internally consistent                                |
| p_recall outside [0, 1]                           | 0                   | Good — valid probability range                              |
| practice_time dtype                               | object → datetime64 | Converted; 0 values lost                                    |

Important Column Reference

| Column              | Dataset                   | What it means                                             |
|---------------------|---------------------------|-----------------------------------------------------------|
| lag_days            | traces_cleaned            | Continuous days since the word was last practised         |
| lag_bin             | traces / forgetting_curve | lag_days grouped into 9 ordinal bins, <1h to 3+ months    |
| history_seen        | traces_cleaned            | Times this learner has seen this word before this session |
| avg_prior_exposures | word_difficulty_cleaned   | Average exposures across all                              |

# Baseline Metrics & KPI Tree

Before slicing by lag, exposure, word, POS, language or course, these are the platform-wide numbers every later percentage is measured against.

499,436  
Total Learning Traces

75,294  
Unique Learners

908,795  
Total Recall Attempts

822,036  
Total Correct Recalls

90.45%  
Overall Word Recall Rate

6,211  
Unique words scored

6  
Learning languages

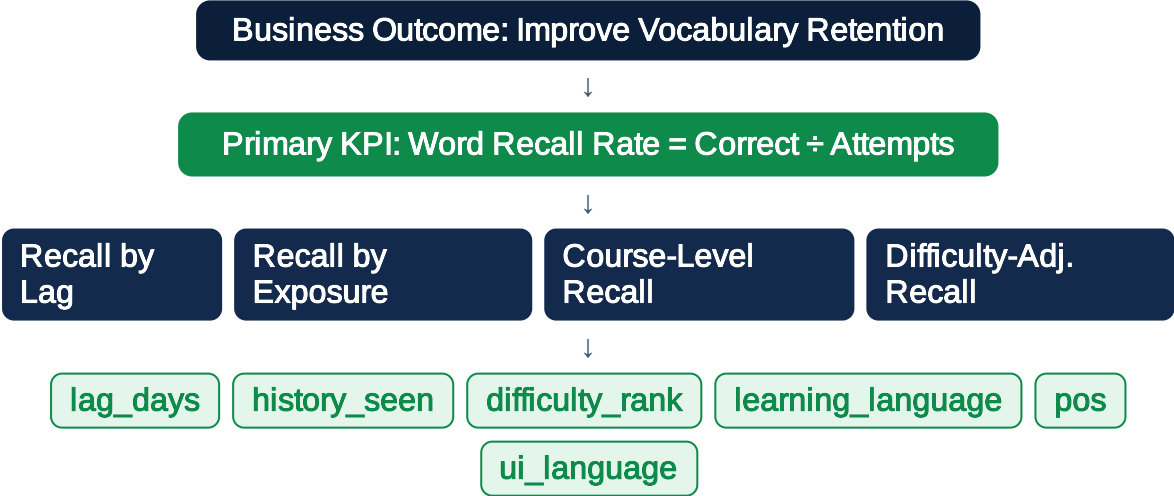
4  
UI languages

8  
Course pairings

0.90d  
Median lag (75th pct 5.13d)

21.8  
Avg prior exposures / word

KPI Tree — From Business Outcome to Drivers



Largest POS Groups (by trace volume)

| Part of Speech | Traces  |
|----------------|---------|
| Noun           | 212,842 |
| Lexical verb   | 86,828  |
| Determiner     | 52,081  |
| Pronoun        | 41,106  |
| Adjective      | 27,595  |

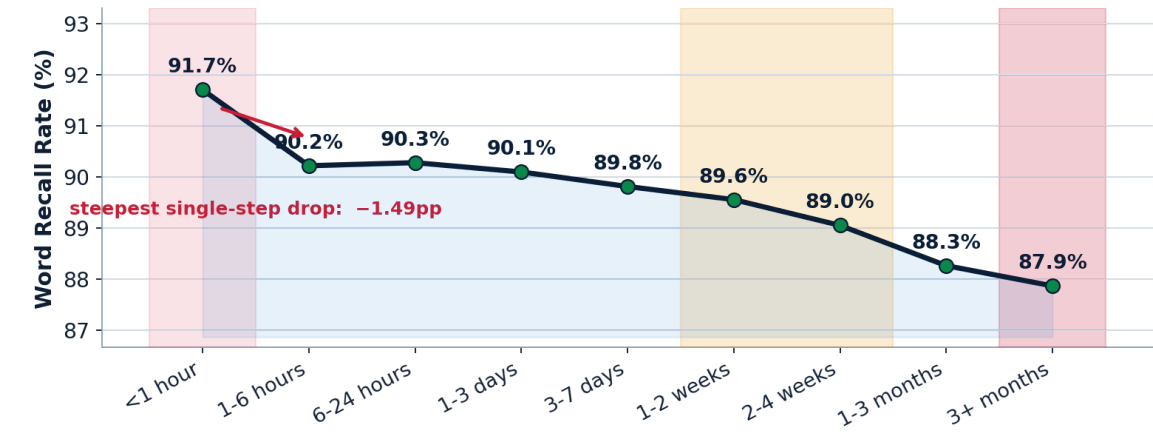
READING THE TREE

Word Recall Rate is driven by three verified levers — time since last practice, prior exposure, and word/course/POS characteristics. Every branch maps to a tested hypothesis (Section 09).

# Time-Based Recall Decay — The Forgetting Curve

**lag\_days** is the continuous number of days since a word was last practised; **lag\_bin** groups it into 9 ordinal buckets so the memory curve is readable. H1 tests whether recall falls as this gap grows. ✓ Strongly Supported

Word Recall Rate by Time Since Last Practice



| Lag bin    | Recall | Step drop | Cum. loss |
|------------|--------|-----------|-----------|
| <1 hour    | 91.70% |           | 0.00pp    |
| 1-6 hours  | 90.21% | -1.49pp   | -1.49pp   |
| 6-24 hours | 90.27% | +0.06pp   | -1.43pp   |
| 1-3 days   | 90.09% | -0.18pp   | -1.61pp   |
| 3-7 days   | 89.81% | -0.28pp   | -1.89pp   |
| 1-2 weeks  | 89.55% | -0.26pp   | -2.15pp   |
| 2-4 weeks  | 89.05% | -0.50pp   | -2.65pp   |
| 1-3 months | 88.26% | -0.79pp   | -3.44pp   |
| 3+ months  | 87.86% | -0.40pp   | -3.84pp   |

## WHERE RECALL FIRST BREACHES 90%

Recall drops below the 90% acceptable threshold at **3-7 days** and keeps falling through 3+ months (87.86%). The steepest single-step loss happens in the first 6 hours (-1.49pp) — the most dangerous review window is immediately after learning, not months later.

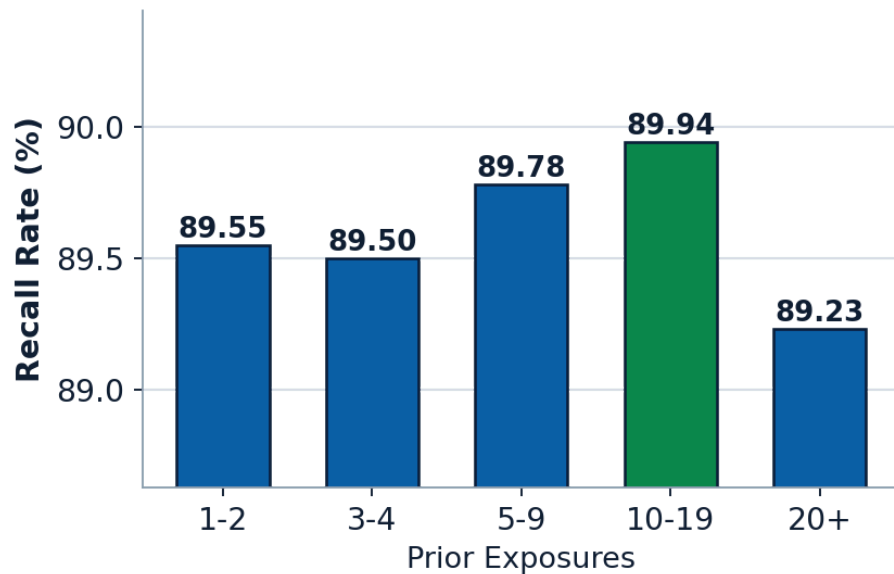
## BUSINESS IMPLICATION

A short early-review nudge within the first few hours would intercept the steepest part of the curve. Words unreviewed for 1+ months should be flagged for priority scheduling regardless of exposure history.

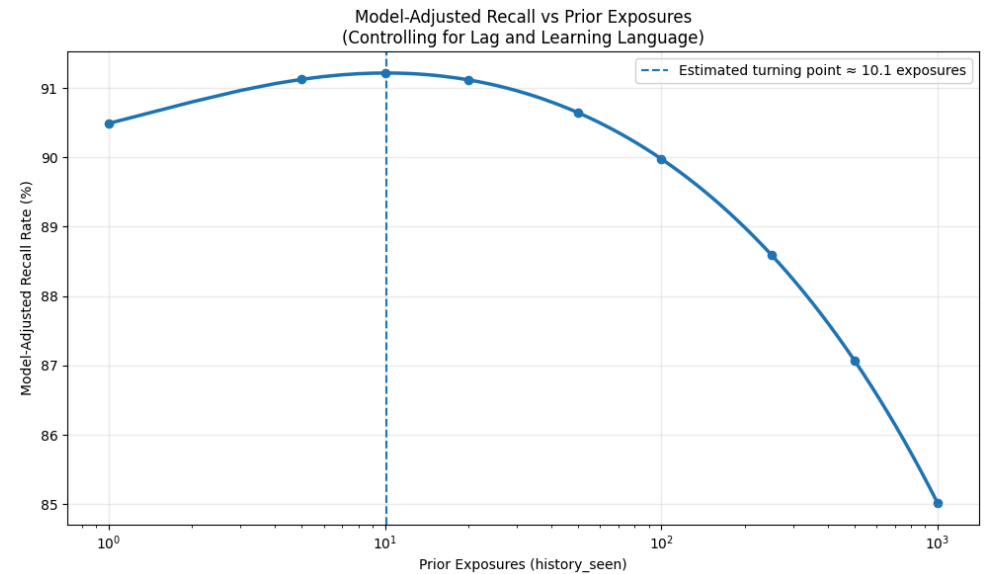
## Exposure Analysis — Does More Practice Improve Recall?

`history_seen` counts how many times a learner has seen a word before; `avg_prior_exposures` averages this across all learners of a word. The intuitive assumption is more repetition = better recall — the data tells a more layered story. ~ Partially Supported

### H2 — Raw Recall by Prior-Exposure Bucket



### H4 — Model-Adjusted Recall vs Prior Exposures (controlling for lag & language)



### CRITICAL FINDING — SURVIVORSHIP BIAS

Raw recall rises from 89.55% at 1–2 exposures to 89.94% at 10–19, then falls to 89.23% at 20+. Words with the most exposures are the *hardest* words — they accumulate repetition precisely because they keep failing. Kruskal-Wallis confirms the bins differ significantly ( $H = 94.39$ ,  $p = 1.53e-19$ ).

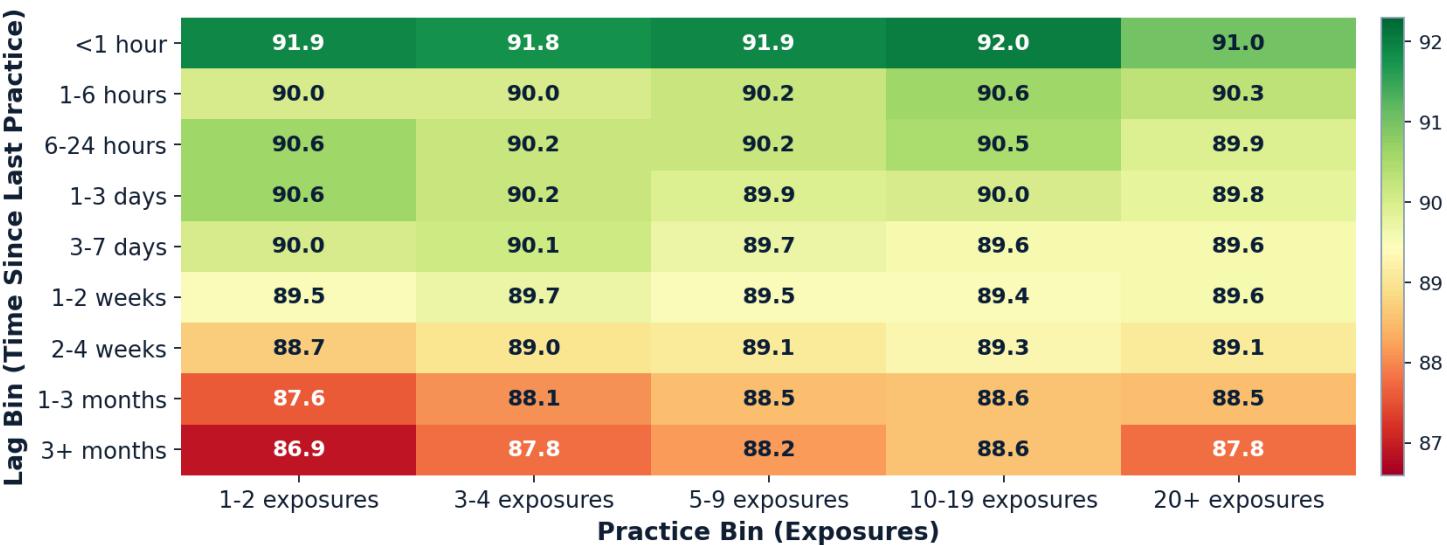
### BUSINESS IMPLICATION

After controlling for lag and language, the true turning point is  $\approx 10.1$  exposures (H4 — supported). Raw exposure count should not be read as a mastery signal beyond this point; a rolling recall-accuracy metric is a more reliable scheduling input than a simple repetition counter.

# Lag × Exposure Interaction

Does prior practice history buffer the damage of a long gap? This is the single most operationally important cut in the dataset — it tells the scheduler which words are safe to leave alone and which need urgent review. ✓ Supported

Word Recall Rate: Lag Bin × Exposure Bucket (item\_recall\_rate, %)



| Comparison              | 1–2 exposures | 10–19 exposures | Buffer |
|-------------------------|---------------|-----------------|--------|
| Recall at <1 hour lag   | 91.9%         | 92.0%           | +0.1pp |
| Recall at 3+ months lag | 86.9%         | 88.6%           | +1.7pp |

LOW-EXPOSURE + LONG-LAG VS. HIGH-EXPOSURE + LONG-LAG

At <1 hour lag, exposure count barely matters (0.1pp gap). At 3+ months, having 10–19 prior exposures instead of just 1–2 is worth **1.7 percentage points** of protection. Practice history only pays off once the gap gets long.

SCHEDULING PRIORITY RULE

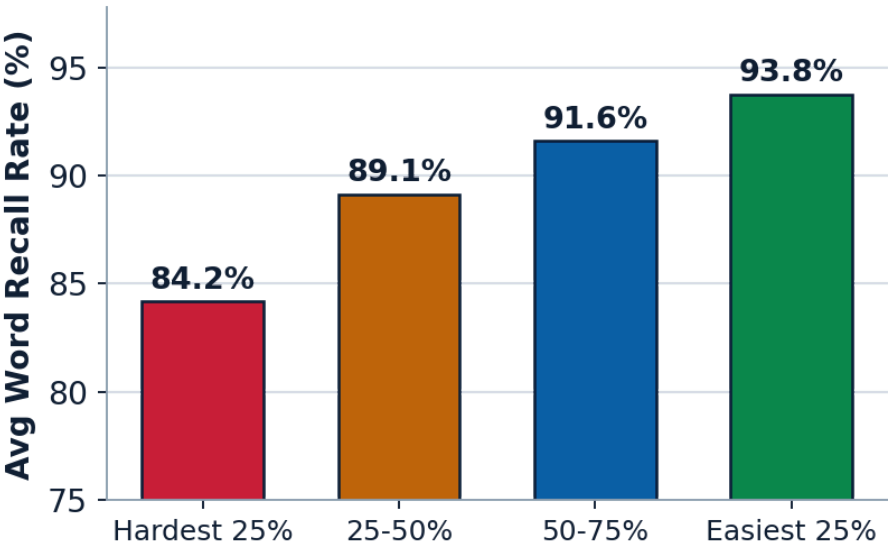
Words combining **long lag AND low exposure** are the highest-risk segment in the entire dataset (recall as low as 86.9%) and should receive the highest review urgency. Words with high exposure survive long gaps far better — they can safely wait longer before the next review.



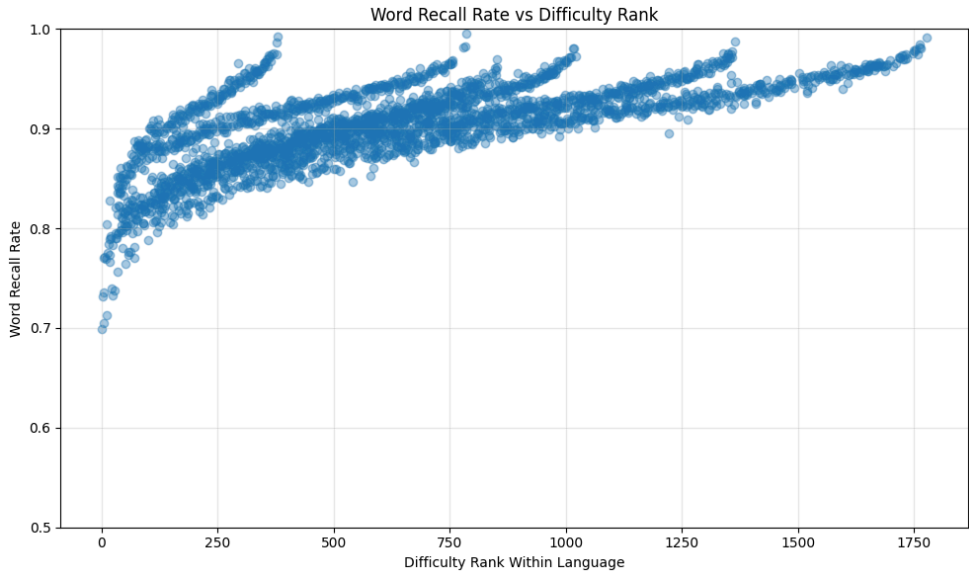
# Word Difficulty — The Strongest Predictor of Recall

difficulty\_rank\_in\_language ranks each word's retrievability within its language. The notebook confirms empirically that **lower rank numbers are the hardest words** (the 20 hardest words all carry rank < 75) — not the easiest. ✓ Strongly Supported

Recall by Difficulty Quartile



Difficulty Rank vs Recall Rate — all 6,211 words



## 10 Hardest Words

| Lang | Word     | POS   | Traces | Recall | Avg exp. |
|------|----------|-------|--------|--------|----------|
| fr   | en       | prn   | 544    | 69.9%  | 5.8      |
| en   | drink    | vblex | 506    | 70.5%  | 86.7     |
| en   | used     | vblex | 1,215  | 71.3%  | 91.4     |
| pt   | refeição | n     | 707    | 73.2%  | 5.7      |
| en   | be       | vbser | 974    | 73.2%  | 119.7    |
| es   | vez      | n     | 1,862  | 73.5%  | 6.0      |
| en   | lot      | n     | 1,154  | 73.8%  | 119.1    |
| en   | were     | vbser | 1,789  | 74.0%  | 127.8    |

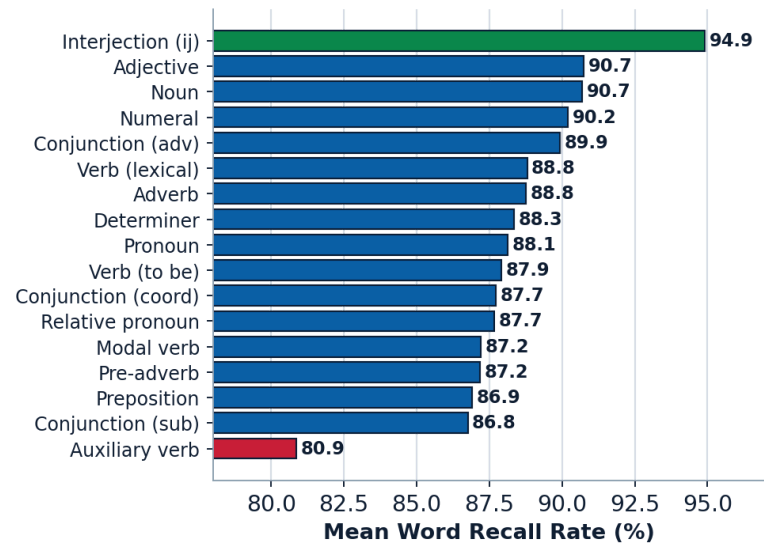
## 10 Easiest Words

| Lang | Word       | POS   | Traces | Recall | Avg exp. |
|------|------------|-------|--------|--------|----------|
| it   | buongiorno | ij    | 689    | 99.6%  | 1.7      |
| pt   | olá        | ij    | 1,224  | 99.3%  | 33.9     |
| en   | adventure  | n     | 570    | 99.1%  | 4.7      |
| es   | estudio    | vblex | 707    | 98.8%  | 6.6      |
| pt   | banana     | n     | 858    | 98.7%  | 4.1      |
| en   | second     | n     | 1,537  | 98.4%  | 5.5      |
| it   | ciao       | ij    | 2,685  | 98.3%  | 6.0      |
| it   | lana       | n     | 629    | 98.2%  | 3.9      |

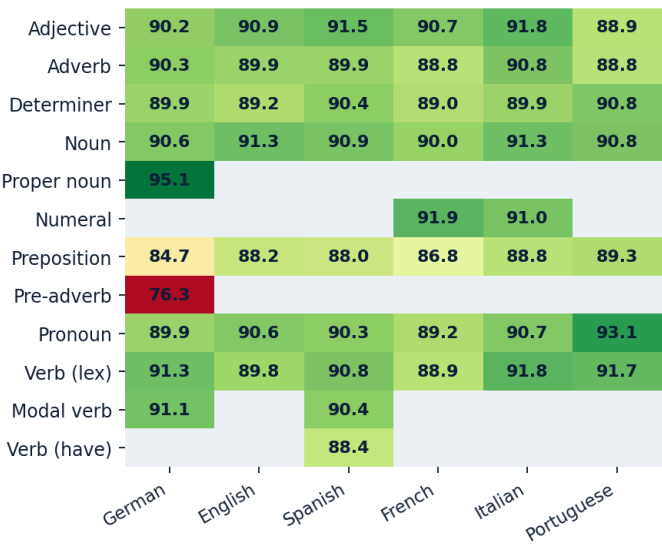
# Part-of-Speech Analysis — Grammar Categories vs Recall

Are nouns easier than adjectives? Are function words harder to hold onto than content words? H8 tests POS alone; H10 tests whether the POS effect changes by language. ✓ Both Strongly Supported

H8 — Recall by Part of Speech (core POS, min. 10 words)



H10 — POS × Language Recall (% , min. 20 words per cell)



## FUNCTION WORDS ARE THE WEAK LINK

Auxiliary verbs recall lowest at 80.9%, followed by prepositions (86.9%) and conjunctions/pronouns/determiners in the 87–88% band — all well below nouns and adjectives (~90.7%). Kruskal-Wallis confirms the spread is real ( $H = 420.19, p < 0.0001$ ).

## H10 — WORST SPECIFIC COMBINATION

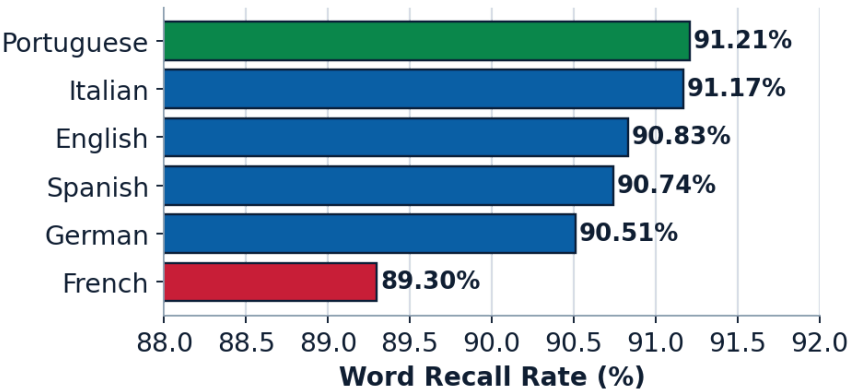
German × Pre-adverb is the single worst POS–language cell at 76.27%. German shows the widest within-language POS spread (18.9pp); prepositions vary most across languages (4.6pp range).

SECTION 10 · H6, H9 & H11

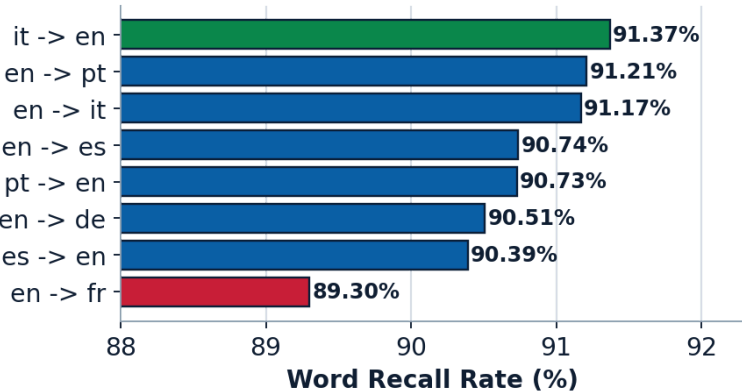
# Language & Course Analysis

Does the target language change recall, and by how much? H6 tests the main effect, H9 tests whether decay speed differs by language, H11 pins down a language-specific 90% threshold. ✓ All Three Supported

H6 — Recall by Learning Language



Recall by Course Pairing (UI → Learning Language)



H11 — When Does Each Language Cross Below 90% Recall?

| Language   | Start recall (<1h) | End recall (3+ mo) | Total decay | Drops below 90% at |
|------------|--------------------|--------------------|-------------|--------------------|
| French     | 90.28%             | 86.70%             | 3.57pp      | 1-6 hours          |
| German     | 92.24%             | 87.48%             | 4.76pp      | 3-7 days           |
| English    | 91.89%             | 88.41%             | 3.47pp      | 1-2 weeks          |
| Spanish    | 92.04%             | 88.06%             | 3.98pp      | 1-2 weeks          |
| Italian    | 92.25%             | 89.40%             | 2.85pp      | 2-4 weeks          |
| Portuguese | 92.70%             | 88.06%             | 4.64pp      | 2-4 weeks          |

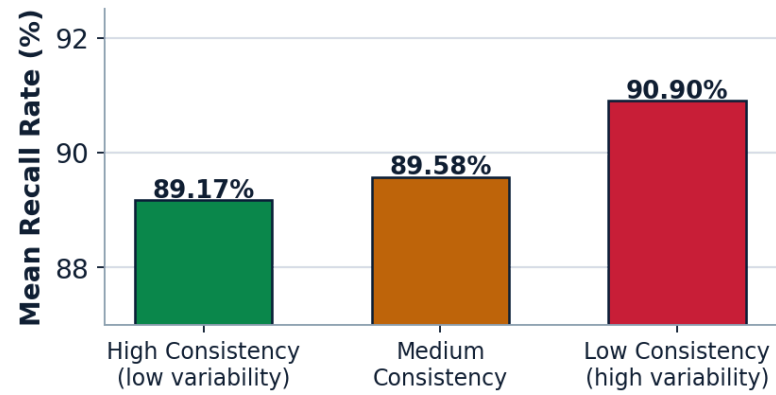
FRENCH IS A STRUCTURAL OUTLIER

French shows the lowest course recall (89.30%), the earliest 90%-threshold breach (within 1–6 hours), and the largest total decay. This points to genuine linguistic complexity, not just a scheduling gap — French needs a dedicated review-interval and content strategy, not a parameter tweak.

Learner Behaviour — Practice Frequency & Consistency

Do learners with steadier practice intervals recall better? cv\_lag (coefficient of variation of a learner's lag) measures consistency; users are also split into frequency buckets by their average lag. ~ Counterintuitive — Survivorship Explained

H7 — Recall by Practice-Consistency Tercile (28,560 users, 5+ sessions each)



| Tier                      | Users | Mean recall | Mean lag | Mean CV |
|---------------------------|-------|-------------|----------|---------|
| High Consistency (low cv) | 9,520 | 89.17%      | 12.36d   | 0.69    |
| Medium Consistency        | 9,520 | 89.58%      | 8.42d    | 1.30    |
| Low Consistency (high cv) | 9,520 | 90.90%      | 5.00d    | 2.25    |

Kruskal-Wallis H = 63.83, p < 0.0001 — tiers differ significantly.

Recall by Practice Frequency × Consistency (within-bucket comparison)

| Frequency bucket     | Consistency  | Users | Mean recall | Mean lag |
|----------------------|--------------|-------|-------------|----------|
| Frequent (<3.4d avg) | Consistent   | 4,760 | 90.23%      | 0.85d    |
| Frequent (<3.4d avg) | Inconsistent | 4,760 | 91.26%      | 0.74d    |
| Moderate (3.4-8.4d)  | Consistent   | 4,760 | 89.58%      | 3.59d    |
| Moderate (3.4-8.4d)  | Inconsistent | 4,760 | 90.37%      | 3.53d    |
| Infrequent (8.4d+)   | Consistent   | 4,760 | 88.08%      | 25.74d   |
| Infrequent (8.4d+)   | Inconsistent | 4,760 | 89.79%      | 17.13d   |

WHY “LESS CONSISTENT” USERS SHOW HIGHER RECALL

Low-consistency (high cv\_lag) learners recall higher (90.90%) than high-consistency learners (89.17%) — this is survivorship, not a real behavioural effect. Irregular reviewers who return after a long gap and still answer correctly are, by definition, selecting for memorable, easy words. The data does not show irregular practice is better; it shows easy words survive irregular practice while hard words drop out.

# Hypothesis Testing — Summary of All 11 Hypotheses

Every hypothesis was tested with the statistical method that fit the data — Pearson/Spearman correlation, Kruskal-Wallis, chi-square, and quadratic regression — never applied just for the sake of it. Full detail for each is in Sections 05–11.

| ID  | Statement (H <sub>1</sub> )                  | Result             | Key statistic                                                         | Priority |
|-----|----------------------------------------------|--------------------|-----------------------------------------------------------------------|----------|
| H1  | Longer lag → lower recall                    | ✓ Supported        | 91.7% → 87.9%; −1.49pp in first 6h                                    | Highest  |
| H2  | More exposures → higher recall               | ~ Partial          | Peaks at 10–19 exposures (89.94%); survivorship confirmed             | Highest  |
| H3  | Higher difficulty rank → lower recall        | ✓ Supported        | r = 0.739; 9.6pp hardest–easiest gap                                  | Highest  |
| H4  | Diminishing returns on exposure (inverted-U) | ✓ Supported        | Model-adjusted turning point ≈ 10.1 exposures                         | High     |
| H5  | Lag × exposure interact                      | ✓ Supported        | 1.7pp protective buffer at 3+ months lag                              | High     |
| H6  | Recall varies by learning language           | ✓ Supported        | French 89.30% vs Portuguese 91.21%                                    | Highest  |
| H7  | Consistent practice → higher recall          | ~ Counterintuitive | Low-consistency 90.90% > High-consistency 89.17% (survivorship)       | Monitor  |
| H8  | POS → differential recall                    | ✓ Supported        | 14.1pp range: interjections 94.9% vs auxiliary verbs 80.9%            | High     |
| H9  | Decay rate differs by language               | ✓ Supported        | French breaches 90% within 1–6h; Italian/Portuguese hold to 2–4 weeks | Highest  |
| H10 | POS × language interaction                   | ✓ Supported        | German×Pre-adverb lowest at 76.3%                                     | High     |
| H11 | Language-specific 90% lag threshold exists   | ✓ Supported        | Thresholds range 0.15 days (French) to 21 days (Italian/Portuguese)   | Highest  |

## READING THIS TABLE

Nine of eleven hypotheses were directly supported. H2 is *partially* supported — the relationship is real but non-linear and contaminated by survivorship bias. H7 is genuinely counterintuitive: the apparent “benefit” of inconsistent practice is a selection effect, not a scheduling insight, and should not be acted on directly.

## SECTION 13 & 14

# Insights & Recommendations

Observation → Evidence → Business Implication

| Observation                                                       | Evidence                                                                       | Business implication                                                                                         |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Recall decays fastest in the first 6 hours after learning         | 91.7% → 90.2% (-1.49pp), the single steepest step in the curve (H1)            | An early review nudge within hours — not days — would intercept the largest source of forgetting.            |
| Raw exposure count is not a mastery signal past ≈10 repetitions   | Model-adjusted recall peaks at 10.1 exposures then declines (H2/H4)            | Replace repetition counters with a rolling recall-accuracy signal for scheduling decisions.                  |
| Word difficulty is the strongest lever available                  | $r = 0.739$ ; 9.6pp gap between hardest and easiest quartiles (H3)             | Difficulty-adaptive review intervals are the most evidence-based structural change available.                |
| Practice history buffers long gaps, but only once the gap is long | +1.7pp recall at 3+ months lag for 10–19 vs 1–2 exposures (H5)                 | Scheduler should combine lag AND exposure signals, prioritising long-lag + low-exposure words first.         |
| French is a structural, not just a scheduling, outlier            | Lowest course recall (89.30%), earliest 90% breach of any language (H6/H9/H11) | French needs its own review-interval defaults and content strategy, not a shared parameter.                  |
| Function words are systematically harder than content words       | Auxiliary verbs 80.9% vs nouns/adjectives ~90.7% (H8/H10)                      | Grammar-aware review formats (not just more drilling) are needed for prepositions, determiners, auxiliaries. |

## Five Scheduling Recommendations

- 1 Language-specific initial review intervals**  
French shortest, Italian/Portuguese longest — parameters read directly off H11. **KPI to watch: Recall at 1–6h lag, by language**
- 2 Difficulty-adaptive scheduling for the hardest quartile**  
Shorter intervals + format-change trigger past 10 exposures with recall < 85%. **KPI to watch: item\_recall\_rate for hardest-quartile words**
- 3 Grammar-contextual exercises for function words**  
Replace pure drilling with contextual practice for prepositions, determiners, auxiliaries. **KPI to watch: Recall by POS, per language, quarterly**
- 4 Rolling recall accuracy as the mastery signal**  
Retire raw exposure count as a scheduling trigger past ≈10 repetitions. **KPI to watch: Correct recalls per session, high-exposure words**
- 5 Lapse re-engagement trigger at 14 days**  
Push notification before words cross into the 1-month decline zone. **KPI to watch: % of words in the 1–3 month / 3+ month lag bins**

SECTION 15

# Executive Dashboard

90.45%

Word Recall Rate

87.9%

Recall at 3+ Months

75,294

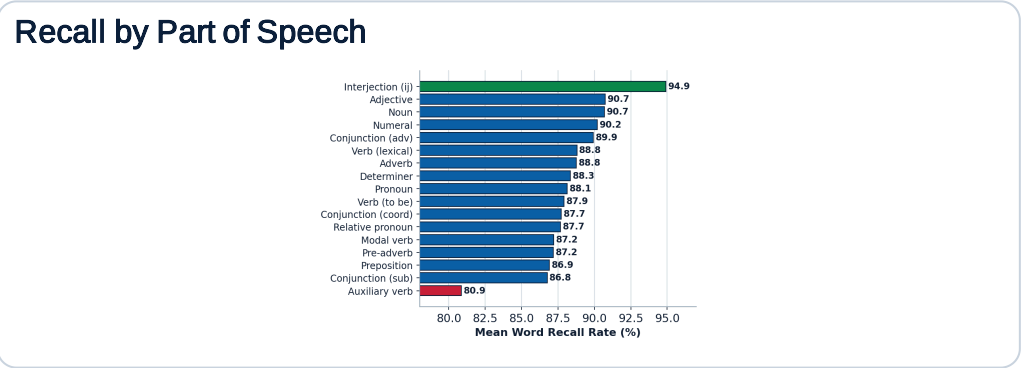
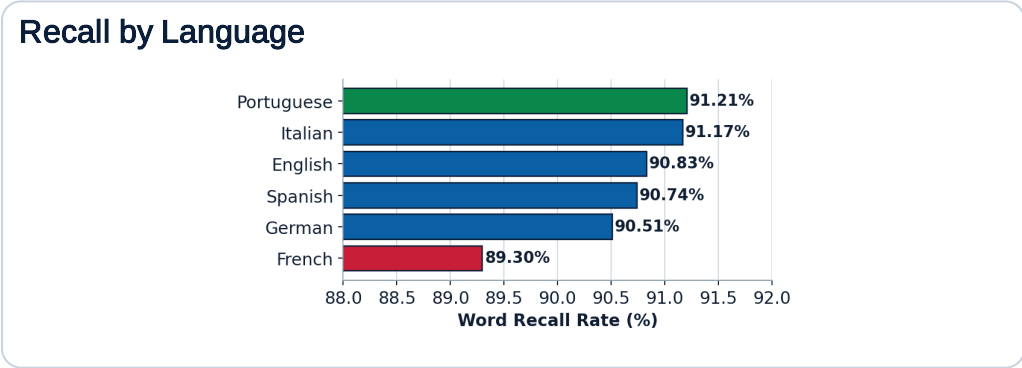
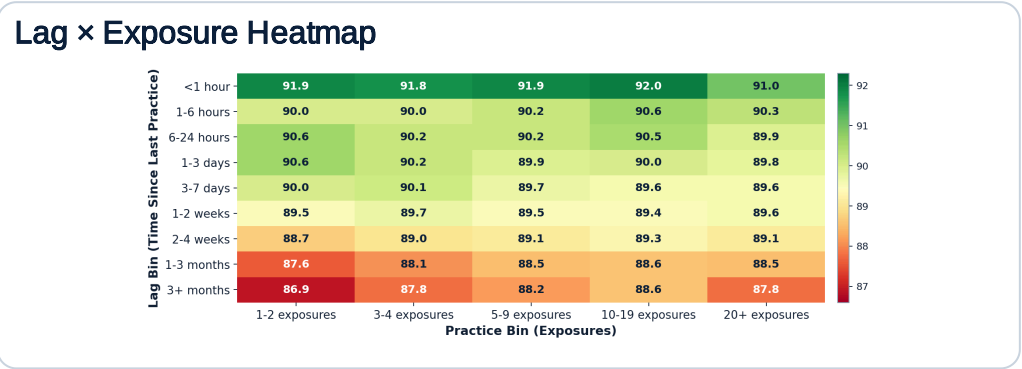
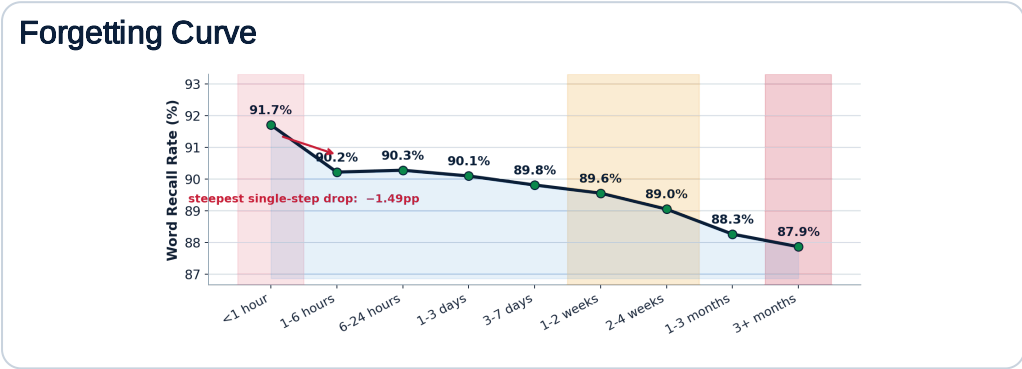
Learners

8

Courses

0.74

Difficulty–Recall r



#### Implementation Roadmap

0–30 DAYS

Set language-specific review intervals from H11. Deploy 14-day lapse push notifications.

30–90 DAYS

A/B test difficulty-adaptive scheduling. Commission grammar-contextual content for weak POS categories.

90+ DAYS

Combine lag + difficulty + language into one scheduling model. Monitor danger-zone word share quarterly.